

Spinning *CFT* Correlators from *AdS* Twistors

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Pre-History: Amplitudes

- Feynman diagrams are hard, long and obscure the physics
- **Revolution E. Witten (2003)** *Perturbative Gauge Theory as a String Theory in Twistor Space*
⇒ Amplitudes localise on simple geometric objects ⇒ *hidden simplicity becomes manifest*
- **N. Arkani-Hamed et al. (2009):**
 - *The S-Matrix in Twistor Space*
 - *A Duality for the S-Matrix*⇒ Dual conformal symmetry & new structures in amplitudes
- **Grassmannian formulation:**
Fourier transform of twistor variables ⇒ amplitudes as contour integrals on $\text{Gr}(k, n)$

Twistor/Grassmannian Geometry replaces Feynman diagrams

(A)dS Correlators: Motivation

- **Direct computation is hard:**

Witten diagrams are technically even more involved and quickly become unwieldy

- **Symmetry is powerful but hidden:**

(S)CFT imposes strong constraints, yet:

- How do we make conformal symmetry *manifest*?
- How do we efficiently handle *conserved operators*?

- **Existing frameworks:**

- Position space • Momentum space
- Embedding space • Mellin space
- Cosmological spinor-helicity
- Analytic / harmonic superspace

Can we find a language where symmetry *and* dynamics are simple?

Main Take Away

- **Kinematics:**

Conformal symmetry + conservation laws become *trivial* in twistor space.

- **Dynamics:**

Nontrivial physics reduces to a *choice of basis* in twistor space.

Kinematics is automatic, dynamics is algebraic.

Some references

- D. Baumann et al., *A New Twist on Spinning (A)dS Correlators* (2024)
- M. Carrillo González et al., *Spinning Conformal Correlators from AdS_4 Twistors* (2025)
- A. Bala et al., *3d Conformal field theory in twistor space* (2025)
- A. Bala et al., *A Supertwistor Formalism for $\mathcal{N} = 1, 2, 3, 4$ SCFT₃* (2025)
- A. Bala et al., *An Ode to the Penrose and Witten transforms in twistor space for 3D CFT* (2025)
- A. Ansari et al. *AdS_4 Boundary Wightman functions in Twistor Space: Factorization, Conformal blocks and a Double Copy* (2025)
- M. Arundine et al. *The Cosmological Grassmannian* (2026)
- S. De *The Vasiliev Grassmannian* (2026)
- A. Bala *Super-Grassmannians for $\mathcal{N} = 2$ to 4 SCFT₃: From AdS_4 Correlators to $\mathcal{N} = 4$ SYM scattering Amplitudes* (2026)
- Y. Huang *Beyond Discontinuities: Cosomological WFCs from the Supersymmetric Orthogonal Grassmannian* (2026)
- and more...

Overview

1. Twistors: From 4d Geometry to 3d Kinematics

2. From Twistors to the Grassmannian

3. Spin is Easy

4. Some enhancements

4.1 Ward Identity

4.2 Bulk Origin of Twistor Correlators

4.3 Spacetime Signature

4.4 Beyond Conserved Currents for 2-point

4d Twistor Space

$$\begin{array}{ccc} & \mathbb{F} = \mathbb{CP}^1 \times (A)dS_{4\mathbb{C}} & \\ \swarrow & & \searrow \\ \text{PT} \subset \mathbb{CP}^3 \quad (Z^A = (\mu^{\dot{\alpha}}, \lambda_{\alpha})) & & (x^{4d})^{\dot{\alpha}\alpha} \end{array}$$

- **Incidence relation:**

$$\mu^{\dot{\alpha}} = x^{\dot{\alpha}\alpha} \lambda_{\alpha}$$

- **Penrose transform:**

Twistor data $\longleftrightarrow$ space-time fields

- **Key idea:** fields $\leftrightarrow$ cohomology on twistor space

Penrose Transform

$$\oint_{\mathbb{CP}^1} \langle \lambda d\lambda \rangle \lambda_{\alpha_1} \cdots \lambda_{\alpha_{2s}} f(Z) = \phi_{\alpha_1 \cdots \alpha_{2s}}(x)$$

- Homogeneity:

$$f(rZ) = r^{-2s-2} f(Z)$$

- $\implies$ automatically solves equations of motion:

$$\partial^{\dot{\alpha}\alpha_1} \phi_{\alpha_1 \cdots} = 0$$

Conformal Symmetry in Twistor Space

- Z^A transforms linearly under $SL(4, \mathbb{C})$

- Generators:

$$T^A_B = Z^A \frac{\partial}{\partial Z^B} \Rightarrow \text{linear action}$$

- Invariants:

$$Z_i \cdot W_j$$

Conformal symmetry is manifest

3d Twistors

- $SL(4, \mathbb{C}) \rightarrow Sp(4)$
- Twistors = dual twistors:

$$\Lambda^M = \Omega^{MN} \Lambda_N$$

- Incidence:

$$\Lambda^M|_x = (\pi^a, \mu^a = x^{ab} \pi_b)$$

- Conformal generators:

$$T^M_N = \Lambda^M \frac{\partial}{\partial \Lambda^N}$$

Everything is built from a single object Λ

Why Kinematics Becomes Trivial

- Derivatives map to twistor operators:

$$\partial_{ab} \leftrightarrow \pi_{(a} \frac{\partial}{\partial \mu^{b)}}$$

- Conservation:

$$\partial_{a_1 a_2} J^{a_1 a_2 \dots} \sim \pi^{a_1} \dots \pi^{a_{2s}} \pi_{(a_1} \frac{\partial}{\partial \mu^{a_2)}}$$

- $\Rightarrow$ vanishes since $\pi^a \pi_a = 0$

Conservation is automatic

Kinematics in Twistor Space

$$\underbrace{\langle O_1 \cdots O_n \rangle}_{\text{hard}} = \oint \prod_i \langle \pi_i d\pi_i \rangle \langle l_i \pi_i \rangle^{2s_i} \underbrace{f(\Lambda_i \cdot \Lambda_j)}_{\text{simple}}$$

- Conformal symmetry $\Rightarrow$ manifest
- Conservation $\Rightarrow$ automatic

Kinematics is automatic, dynamics is a choice of f

From Twistors to the Grassmannian

- Twistor amplitudes naturally involve contractions:

$$\Lambda_i \cdot \Lambda_j$$

- Introduce **link variables** c_{ij} :

$$\Lambda_i \cdot \Lambda_j \longleftrightarrow c_{ij}$$

- Perform a (half) Fourier transform:

$$f(\Lambda_i \cdot \Lambda_j) = \int dc_{ij} e^{i c_{ij} Z_i \cdot W_j} A(c_{ij})$$

- $\Rightarrow$ amplitudes become functions of a matrix $C = (c_{ij})$

Grassmannian & Momentum Space

- Momentum conservation:

$$\sum_i \lambda_i^{(a} \bar{\lambda}_i^{b)} = 0$$

- $\Rightarrow$ correlators as integrals over $\text{Gr}(n, 2n)$:

$$\underbrace{\psi_n(p_i)}_{\text{hard}} = \int \frac{d^{n \times 2n} C}{\text{GL}(n)} \delta(C \cdot Q \cdot C^T) \delta(C \cdot \Lambda) \underbrace{A_n(C)}_{\text{simple}}$$

Spin is Easy

- Correlators:

$$\langle O_1 \cdots O_n \rangle = \oint \cdots \langle l_i \pi_i \rangle^{2s_i} f(\Lambda_i \cdot \Lambda_j)$$

- **Spin is encoded algebraically:**

$$\langle l_i \pi_i \rangle^{2s_i}$$

- no tensor structures
 - no differential constraints
-
- Contrast:
 - position space: complicated index contractions
 - momentum space: non-trivial helicity bookkeeping

Spin is carried by simple polynomials

Double Copy

- Correlators are encoded in: $f(\Lambda_i \cdot \Lambda_j)$:

$$f_{\text{scalar}} = \frac{1}{(\Lambda_1 \cdot \Lambda_2)(\Lambda_2 \cdot \Lambda_3)(\Lambda_3 \cdot \Lambda_1)}, \quad f_{\text{YM}} = \frac{\log(\Lambda_2 \cdot \Lambda_3) \log(\Lambda_3 \cdot \Lambda_1)}{(\Lambda_1 \cdot \Lambda_2)^4}$$

- Double copy:**

$$f_{\text{grav}} \sim \frac{f_{\text{YM}} \times f_{\text{YM}}}{f_{\text{scalar}}}$$

- Purely algebraic:
 - same building blocks $(\Lambda_i \cdot \Lambda_j)$
 - no kinematic rearrangements (unlike position/momentum space, where this is non-trivial)

Double copy = algebra of twistor invariants

Ward Identity

- Grassmannian:

$$\psi_n = \int_{\text{Gr}} \delta(\cdots) \underbrace{A_n(C)}_{\text{rational}}$$

$\Rightarrow$ naturally gives the **homogeneous part** of the correlator

- Twistor:

$$\langle O_1 \cdots O_n \rangle = \oint \cdots \underbrace{f(\Lambda_i \cdot \Lambda_j)}_{\text{full answer}}$$

$\Rightarrow$ directly produces the **full position-space correlator**

- Example:

- F^3 : homogeneous $\Rightarrow$ both approaches equivalent
- YM: Grassmannian $\Rightarrow$ discontinuity only

Twistor: full answer vs Grassmannian: homogeneous part

Bulk Origin of Twistor Correlators

- Construct bulk correlators in $(A)dS_4$:

$$\langle F_s(X_1) \bar{F}_s(X_2) \rangle$$

- Take boundary limit:

$$\langle O_s(P_1) O_s(P_2) \rangle = \lim_{z \rightarrow 0} (z_1 z_2)^{-\Delta} \langle F_s \bar{F}_s \rangle$$

- Interpretation:
 - Twistor correlators are not ad hoc
 - They arise from **bulk Penrose transforms**
- Benefit:

Bulk physics $\longleftrightarrow$ Boundary correlators

Twistor correlators have a direct bulk origin

Spacetime Signature

- Witten transform requires split signature
- Twistor construction works in complexified spacetime
- Advantage:
 - impose reality conditions *at the end*
 - $\delta^{[n]}(\Lambda_i \cdot \Lambda_j) \implies$ rational functions + logs

Beyond Conserved Currents for 2-point

- Grassmannian gives a divergent answer in this case
- In twistor space, it can be regularised via Euclidean continuation
- Standard case:

$$\Delta = s + 1 \quad (\text{conserved})$$

- Generalisation:

$$\Delta > s + 1 \quad (\text{non-conserved})$$

- Bulk interpretation:

massive fields in AdS

- Why useful:

- extends twistor methods beyond conserved sector
- captures more general CFT data

Twistor framework extends to general Δ

Conclusions

- In the Grassmannian/twistor formulation, **conformal symmetry and conservation are manifest**
- Introducing spin in the correlators is a matter of raising the representative to the correct power $\implies$ Double Copy manifest at 3 points
- Ward identity is straightforward in Twistor Space
- Bulk origin exists in terms of AdS_4 Twistors
- 2-point is finite and can be generalised to massive fields in AdS